

```

%% Problem 1
clear; clc;

rng(13);

p = 0.35 + (0.95-0.35)*rand(1,40);

answers = rand(5000,40) < p;

scores = sum(answers,2);

avgScore = mean(scores);
stdScore = std(scores);

fprintf('평균 점수 = %.2f\n',avgScore);
fprintf('표준편차 = %.2f\n',stdScore);

histogram(scores);
xlabel('Score');
ylabel('Frequency');
title('Score Distribution');

ratio = mean(scores >= avgScore);

fprintf('평균 이상 학생 비율 = %.4f\n',ratio);

```